Network Security Definitions (Chapter 8)

8.1 What is Network Security?

Confidentiality The property that ensures only the sender and intended receiver can understand the message contents. Achieved via **encryption**.
Authentication The property that allows communicating parties to confirm each other's identity.
Message Integrity The property that ensures a message has not been altered in transit. Achieved via cryptographic hashes or MACs.
Availability The property that ensures network services are accessible to legitimate users. Defends against attacks like Denial of Service (DoS).
Eavesdropping An attack where an intruder intercepts and reads messages, violating confidentiality.
Impersonation (Spoofing) An attack where an intruder fakes the source address or identity of another user.
Playback Attack An attack where an intruder records a legitimate, encrypted session (e.g., a login) and replays it later to impersonate the legitimate user.

8.2 Principles of Cryptography

Plaintext The original, unencrypted message.
Ciphertext The encrypted, scrambled message.
Key A secret piece of information used by an encryption/decryption algorithm.
Symmetric Key Cryptography A cryptographic system where the sender and receiver share a **single, common secret key** for both encryption and decryption. It is fast and efficient.
- **AES (Advanced Encryption Standard):** The modern, dominant symmetric key algorithm.

Public Key (Asymmetric) Cryptography A system where each user has a **key pair**: a **public key** (known to everyone) and a **private key** (known only to the user).
- A message encrypted with the public key can only be decrypted with the corresponding private key. This provides **confidentiality**.

RSA (Rivest, Shamir, Adelson) The most widely used public key algorithm. Its security is based on the difficulty of factoring large numbers.
Session Key A temporary, symmetric key used for encrypting data during a single communication session. It is typically exchanged at the start of a session using slower public key cryptography.

8.3 Message Integrity and Authentication

Challenge-Response Protocol An authentication method used to prevent playback attacks. One party sends an unpredictable value (the "challenge"), and the other party must return a cryptographically transformed version of it (the "response") to prove their identity.
Nonce A "number used once." A random, unpredictable value used as the challenge in a challenge-response protocol.
Man-in-the-Middle (MITM) Attack A sophisticated attack where an intruder places themself in the middle of a communication, impersonating each party to the other, allowing them to read and modify all messages without being detected.
Cryptographic Hash Function A function that takes an arbitrary-length message and produces a fixed-length "fingerprint" called a **message digest**. Key properties include being one-way and collision-resistant. Examples: SHA-256, SHA-3.
Digital Signature A cryptographic technique that provides authentication, integrity, and non-repudiation. It is created by **encrypting a hash of the message with the sender's private key**.
Non-repudiation The property that ensures a sender cannot deny having sent a message that they digitally signed.
Message Authentication Code (MAC) Also known as a keyed hash. It is a message digest created by hashing the message concatenated with a shared secret key. It provides both authentication and integrity, but not non-repudiation.
Certification Authority (CA) A trusted third party that issues **digital certificates**.
Digital Certificate A data structure that binds a public key to a specific entity (e.g., a website). The certificate itself is digitally signed by a CA, allowing others to trust the validity of the public key.

8.4 & 8.5 Securing Email and TCP (TLS)

TLS (Transport Layer Security) The standard security protocol for providing a secure channel over TCP. It is the protocol that underlies **HTTPS**. Successor to SSL.
TLS Handshake The initial phase of a TLS connection where the client and server authenticate each other (or client authenticates server), agree on cryptographic algorithms, and generate a shared symmetric key for the session.
Cipher Suite The specific set of algorithms to be used for a TLS session, including the key exchange method, the bulk encryption algorithm (e.g., AES), and the message authentication code algorithm (e.g., HMAC-SHA256).
TLS 1-RTT Handshake The standard handshake in modern TLS (1.3), where the client sends its key exchange parameters

in the first message, allowing the handshake to complete in one round-trip time.
TLS 0-RTT Handshake An optional mode for resuming a previous TLS session, where the client sends encrypted application data in the very first message. While fast, it is vulnerable to playback attacks.